

Use-Case Flow Specification

Pet Adoption & Shelter Management System — Lab 1

Use Case: UC-06 Finalize Adoption

Actors: Shelter Staff (primary), Payment Gateway (supporting)

Includes: UC-07 Process Adoption Payment

Preconditions: The applicant's questionnaire (UC-04) has been Approved by Shelter Staff, and the target animal's status is 'Ready for Adoption' (not under medical quarantine or an active foster placement).

Postconditions: The animal's status is updated to 'Adopted', the animal is removed from the active search catalogue, the linked questionnaire is closed, and a payment confirmation record is stored against the adoption.

Main Success Scenario

1. Shelter Staff opens the Approved questionnaire for an applicant and selects “Finalize Adoption” for the linked animal.
2. System re-verifies the animal's availability status (UC-03) to confirm it is still 'Ready for Adoption'.
3. System displays the adoption fee amount for the animal.
4. Shelter Staff selects a payment method (credit card, debit card, or digital wallet) on behalf of the applicant.
5. System sends a payment authorization request to the external Payment Gateway (UC-07).
6. Payment Gateway authorizes the payment.
7. System updates the animal's status to 'Adopted' and removes it from the active search catalogue.
8. System closes the linked questionnaire and records the payment confirmation against the adoption.
9. System displays a confirmation message and generates an adoption receipt for the applicant.
10. Use case ends successfully.

Alternate Flow

6a. Payment Declined

- 6a1. System displays a “payment failed” message to Shelter Staff.
- 6a2. Shelter Staff is prompted to select another payment method for the applicant.
- 6a3. If payment fails again (after 2 attempts), the finalization is cancelled, the animal's status remains 'Ready for Adoption', and the applicant is notified.